

ANALYSTLAB AFRICA

Experience Lab Internship Programme



HealthConnect Clinic

ML System Architecture Design

By

Ogheneughwe Godwin AKISE

Project: HealthConnect Clinic Experience Lab – Improving Patient Appointment Attendance and Healthcare Support Using Data and AI

Track: Data Science

Prepared for the AnalystLab Africa Experience Lab

Week 4 Submission – Supporting Document

Table of Contents

- 1. Introduction
- 2. Architecture Overview
- 3. Component Descriptions
- 4. Architecture Diagram
- 5. Technology Stack
- 6. Design Considerations
- References

1. Introduction

This document presents an initial, high-level system architecture for the HealthConnect no-show prediction solution. It translates the machine learning problem defined in the Data Science track's Week 4 Machine Learning Problem Definition Document into a structural view of the components a working system would need, so that the Machine Learning Engineering track can use it as a starting point for detailed system design.

Consistent with the Week 4 brief, this is a planning-stage design: no component described here has been built or deployed. The architecture is deliberately kept technology-agnostic where possible, since the final tooling choices belong to the Machine Learning Engineering track.

2. Architecture Overview

The proposed architecture follows a standard offline-training, online-serving pattern for a tabular classification problem. Raw appointment data flows through validation, preprocessing and feature engineering stages before being used to train and evaluate candidate models. An approved model is versioned in a model registry and used by a prediction service to score new appointments as they are booked. Predictions feed a staff-facing dashboard, and actual appointment outcomes are logged and fed back into the training data to support periodic retraining.

3. Component Descriptions

- **Appointment Data Source:** the HealthConnect Appointment Dataset (currently a CSV export; a live system would likely read from a clinic scheduling database).
- **Data Ingestion & Validation:** loads new appointment records and checks them against the expected schema defined in the Data Dictionary.
- **Data Quality Checks:** automated checks for missingness, invalid values, and potential leakage risks, including a check that fields such as `waiting_time_minutes` are not passed to the model before their status has been confirmed (see Section 6).
- **Preprocessing:** cleaning, encoding of categorical variables and imputation of the small number of missing values identified in Week 4.
- **Feature Engineering:** derives model-ready features such as booking lead time and each patient's historical no-show rate.
- **Train / Validation / Test Split:** a chronological split, consistent with the time-based evaluation approach proposed in the Problem Definition Document.
- **Model Training:** trains and compares candidate models, starting with a logistic regression baseline and progressing to gradient-boosted trees.
- **Model Evaluation:** computes precision, recall, F1-score and ROC-AUC, and includes the subgroup fairness review described in Section 6.
- **Model Registry:** stores approved, versioned models so that any prediction can be traced back to the exact model and data version that produced it.
- **Prediction Service:** generates a no-show risk score for each newly booked appointment, using only features available at booking time.
- **Clinic Staff Risk Dashboard:** surfaces high-risk appointments to clinic staff for prioritised outreach; this is a decision-support view, not an automated action.
- **Outcome Logging & Feedback Loop:** records the actual appointment outcome and feeds it back into the training dataset to support periodic retraining.

4. Architecture Diagram

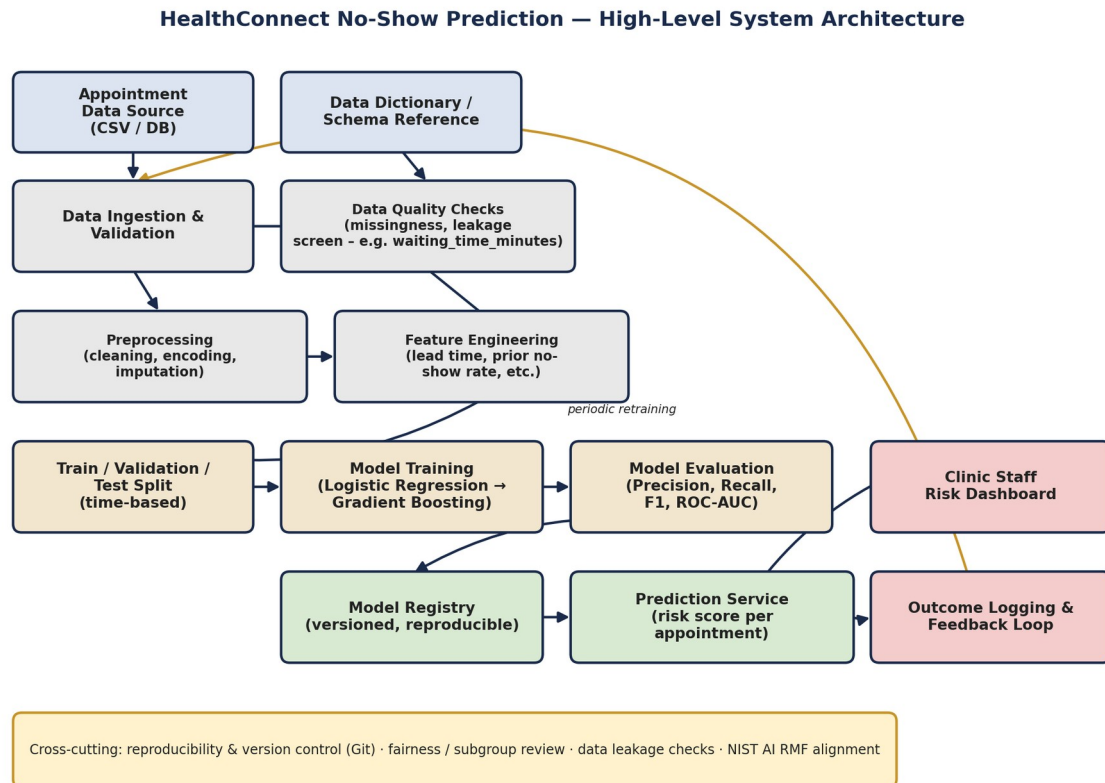


Figure 1. High-level system architecture for the HealthConnect no-show prediction solution.

5. Technology Stack

The following tools are proposed, consistent with the Data Science and Machine Learning Engineering track tooling listed in the Week 4 brief; final selection remains the responsibility of the Machine Learning Engineering track:

- **Data processing and modelling:** Python, Pandas, NumPy and scikit-learn, extending to a gradient boosting library for the candidate ensemble model.
- **Experimentation environment:** Jupyter Notebook or Google Colab for iterative analysis, as used throughout Week 4.
- **Version control:** Git and GitHub, for both code and model versioning.
- **Model registry and serving:** a lightweight file-based or framework-native model registry is proposed at this stage, given the project's scale; a more formal registry could be introduced if the solution progresses beyond the Experience Lab.

6. Design Considerations

The architecture explicitly separates data quality checks from feature engineering so that leakage risks, such as the unresolved status of `waiting_time_minutes` identified in Week 4, can be caught before they reach the training pipeline (Kaufman et al., 2012). The model registry and outcome feedback loop are included specifically to support reproducibility and to guard against the kind of hidden technical debt that commonly accumulates in real-world machine learning systems when versioning and monitoring are treated as an afterthought (Sculley et al., 2015). The subgroup fairness check within Model

Evaluation reflects the responsible-AI considerations set out in the Problem Definition Document (Tabassi, 2023).

References

- AnalystLab Africa. (2026). *AnalystLab Africa Experience Lab – Week 4: HealthConnect project kickoff & problem understanding*. Assignment brief.
- AnalystLab Africa. (2026). *HealthConnect Clinic: Machine learning problem definition document*. Week 4 submission.
- Kaufman, S., Rosset, S., Perlich, C., & Stitelman, O. (2012). *Leakage in data mining: Formulation, detection, and avoidance*. ACM Transactions on Knowledge Discovery from Data, 6(4), Article 15. <https://doi.org/10.1145/2382577.2382579>
- Sculley, D., Holt, G., Golovin, D., Davydov, E., Phillips, T., Ebner, D., Chaudhary, V., Young, M., Crespo, J.-F., & Dennison, D. (2015). *Hidden technical debt in machine learning systems*. In Advances in Neural Information Processing Systems 28 (NeurIPS 2015) (pp. 2503–2511). <https://papers.nips.cc/paper/5656-hidden-technical-debt-in-machine-learning-systems>
- Tabassi, E. (2023). *Artificial intelligence risk management framework (AI RMF 1.0)* (NIST AI 100-1). National Institute of Standards and Technology. <https://www.nist.gov/publications/artificial-intelligence-risk-management-framework-ai-rmf-10>